

PRACTICAL 9

Q1. Write a program to stored the data with the help of read and write operation using RFID module.

Connections:

Raspberry Pi Pin	NFC RFID Module
Pin 6	GND
Pin 2	VCC
Pin 3	SDA
Pin 5	SCL

Channel Number	On/ Off
Channel 1	On
Channel 2	Off

Enter the following Commands:

Command 1: `sudo raspi-config (enable i2c)`

Command 2: `pip3 install adafruit-circuitpython-pn532 --break-system-packages`

Enter the following Python Programs:

Program 1: to verify the card number

```
import board
import busio

from adafruit_pn532.i2c import PN532_I2C

i2c = busio.I2C(board.SCL, board.SDA)

pn532= PN532_I2C(i2c)

ic, ver, rev, support = pn532.firmware_version
```

```

print(f"successs! Found PN532 with Fireware: {ver}.{rev}")

print("Place your blue fob or white card on the red board...")

while True:

    uid= pn532.read_passive_target(timeout=0.5)

    if uid is not None:

        print(f"Found Tag! ID is : {[hex(i) for i in uid]}")

```

Program 2: Write the data

```

import board

import busio

from adafruit_pn532.i2c import PN532_I2C

# Setup I2C

i2c = busio.I2C(board.SCL, board.SDA)

pn532 = PN532_I2C(i2c)

print("--- NFC Writer ---")

text_to_write = input("Enter a small message to save to the card: ")

# Data must be exactly 16 bytes for a single block

# We pad the text with spaces if it's too short

data = text_to_write.ljust(16).encode()

print("Now, place your white card on the reader...")

while True:

    uid = pn532.read_passive_target(timeout=0.5)

    if uid is not None:

        try:

            # We use block 4 (Sector 1) to avoid messing with system blocks

```

```

# This requires a 'default' key for Mifare cards

key = b'\xFF\xFF\xFF\xFF\xFF\xFF'

if pn532.mifare_classic_authenticate_block(uid, 4, 0x60, key):

    pn532.mifare_classic_write_block(4, data)

    print(f"Success! '{text_to_write}' written to card.")

    break

else:

    print("Authentication failed!")

except Exception as e:

    print(f"Error: {e}")

    break

```

Program 3: Read the data

```

import board

import busio

from adafruit_pn532.i2c import PN532_I2C

i2c = busio.I2C(board.SCL, board.SDA)

pn532 = PN532_I2C(i2c)

print("Waiting for card to read stored data...")

while True:

    uid = pn532.read_passive_target(timeout=0.5)

    if uid is not None:

```

```

key = b'\xFF\xFF\xFF\xFF\xFF\xFF'

if pn532.mifare_classic_authenticate_block(uid, 4, 0x60, key):

    data = pn532.mifare_classic_read_block(4)

    if data is not None:

        print(f"Stored Data: {data.decode().strip()}")

        break

```

Q2. Write a program to display unique ID for the input data using RFID module.

Command:

Command 1: sudo raspi-config (enable i2c)

Command 2: sudo reboot

Command 3: pip3 install adafruit-circuitpython-pn532 --break-system-packages

Command 3: sudo apt install -y libnfc-bin libnfc-dev libusb-dev libpcsclite-dev i2c-tools

Command 4: sudo nano /etc/nfc/libnfc.conf

Edit the code: #Allow device auto-detection (default: true)

#Note: if this auto-detection is disabled, user has to set manually a device

#configuration using file or environment variable

allow_autoscan = false

#Allow intrusive auto-detection (default: false)

#Warning: intrusive auto-detection can seriously disturb other devices

#This option is not recommended, user should prefer to add manually his device.

#allow_intrusive_scan = false

#Set log level (default: error)

#Valid log levels are (in order of verbosity): 0 (none), 1 (error), 2 (info), 3 (debug)

#Note: if you compiled with --enable-debug option, the default log level is "debug"

#log_level = 1

#Manually set default device (no default)

#To set a default device, you must set both name and connstring for your device

#Note: if autoscan is enabled, default device will be the first device available in device list.

```
device.name = "PN532 over I2C"
```

```
device.connstring = "pn532_i2c:/dev/i2c-1"
```

Save the file.

Command 5: `i2cdetect -y 1` (optional put `sudo`)

Command 6: `nfc-list`

Enter the following Python program in thonny.

```
import subprocess
```

```
import time
```

```
NAME = "Upasana" # Put your name here
```

```
last_uid = None
```

```
try:
```

```
    while True:
```

```
        output = subprocess.getoutput("nfc-list")
```

```
        if "UID" in output:
```

```
            for line in output.splitlines():
```

```
                if "UID" in line:
```

```
                    uid = line.split(":")[1].strip().replace(" ", "")
```

```
                    if uid != last_uid:
```

```
                        print(f"{NAME}: {uid}")
```

```
                        last_uid = uid
```

```
                    break
```

```
        time.sleep(1)
except KeyboardInterrupt:
    print("\nStopped")
```